Problem

For the circuit in Fig. 3.47, $v1$ and $v2$ are related as:

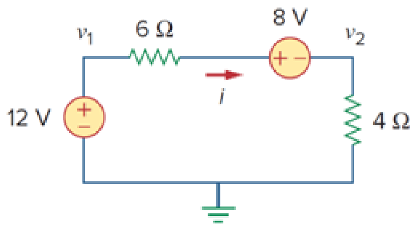
(a) $v1 = 6i + 8 + v2$

(b) $v1 = 6i - 8 + v2$

(c) $v1 = -6i + 8 + v2$

(d) $v1 = -6i - 8 + v2$

Figure 3.47



Step-by-step solution

Step 1 of 3

Consider the following circuit:

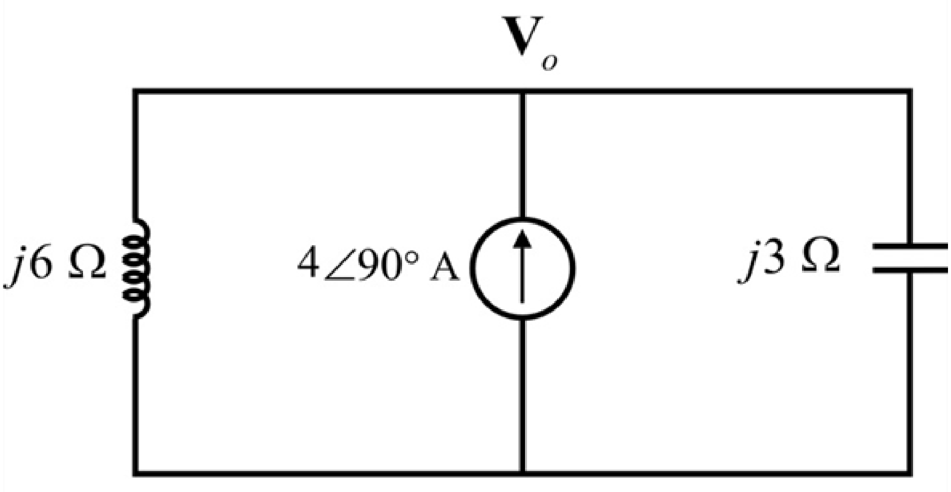


Figure 1

Step 2 of 3

Apply Kirchhoff's current law at the node of voltage $\mathbf{V}_o$.

$$\frac{\mathbf{V}_o}{j6} - 4\angle 90° + \frac{\mathbf{V}_o}{-j3} = 0$$

$$\mathbf{V}_o\left(\frac{1}{j6} - \frac{1}{j3}\right) = 4\angle 90°$$

$$\mathbf{V}_o\left(\frac{-j3}{(j6)(j3)}\right) = j4$$

$$\mathbf{V}_o\left(\frac{-1}{j6}\right) = j4$$

Simplify further,

$$\begin{aligned}\mathbf{V}_o &= -(j4)(j6) \\ &= 24 \text{ V}\end{aligned}$$

Step 3 of 3

Therefore, the value of voltage $\mathbf{V}_o$ is 24 V.

Hence, the correct option is **(d).**